

Everest Engineering College

Sanepa, Lalitpur

Assignment 3 , 2024

Level : BE First Semester

Subject: Calculus I

Due Date: Jan 12, 2025

Students are required to give their answers in their own words as far as practicable and copy from others is strictly Prohibited.

Attempt all the Questions:

1. Find the asymptotes to the following curves
 - a. $(y - a)^2(x^2 - a^2) = x^4 + a^4$
 - b. $x^3 - 2y^2 - 2x^2y - xy^2 + xy - y^2 + 1 = 0$
 - c. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 - d. $4x^3 - 3xy^2 - y^3 + 2x^2 - xy - y^2 - 1 = 0$
 - e. $x^2(x - y)^2 - a^2(x^2 + y^2) = 0$.
2. Trace the following curves:
 - a. $x^2(a^2 + y^2) = y^2(a^2 - y^2)$ [**Hint: example 38, page 85**]
 - b. $x^2y^2 = x^2 - 1$ [**Hint: see similar example 37, page 84**]
 - c. $y^2 = x^3$ [**Hint: see similar example 34, page 82**]
 - e. $x^{2/3} + y^{2/3} = a^{2/3}$ [Hint: example 33, page 81]
3. Plot the parametric equation $x = t^2 - 2t$, and $y = t + 1$ for $0 \leq t \leq 4$.
4. Find the radius of curvature at any point (x,y) of the Cartesian curve
 - a. $y = a \cosh(\frac{x}{a})$
 - b. $xy = c^2$
5. Find the radius of curvature of the curve $y^2 = 4x$ at the vertex (0,0).
6. Find the radius of curvature for the parametric curve at the points indicated.
 - a. $x = a \cos \theta$, $y = a \sin \theta$ at (x, y) .
 - b. $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ at $\theta = \pi/4$.
 - c. $x = a(t + \sin t)$, $y = a(1 - \cos t)$ at (x, y) .
7. Find the radius of curvature of the following polar curves at the points indicated.
 - a. $r = a(1 - \cos \theta)$ at (r, θ) .
 - b. $r^2 \cos 2\theta = a^2$ at (r, θ) .
 - c. $r = a(\theta + \sin \theta)$ at $\theta = 0$.
8. Find the radius curvature at origin of the following curves.
 - a. $3x^2 + 4y^2 = 2x$
 - b. $4x^2 - 3xy + y^2 - 3y = 0$

c. $x^3 + y^3 = 3axy$

d. $x^4 + y^2 = 6a(x + y)$.

9. Find the radius of curvature of the curve $y = x^2(x - 3)$ at the points where the tangent is parallel to x-axis.

10. Show that radius of curvature at any point of a circle $x^2 + y^2 = a^2$ is constant and its curvature at any point is also constant.

[Show that $\rho = a$ and $\kappa = 1/a$]

11. Evaluate the following integral

a. $\int \frac{1}{\sqrt{2ax-x^2}} dx$

b. $\int \frac{1}{x[10+7\log x+(\log x)^2]} dx$

c. $\int \frac{3e^{2x}+3e^{4x}}{e^x+e^{-x}} dx$

d. $\int \frac{x^2+x+1}{\sqrt{1-x^2}} dx$

[Hint: $\int \frac{x^2+x+1}{\sqrt{1-x^2}} dx = -\int \frac{-x^2-x-1}{\sqrt{1-x^2}} dx = -\int \frac{(1-x^2)-x-2}{\sqrt{1-x^2}} dx = -\int (\frac{1-x^2}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} - \frac{2}{\sqrt{1-x^2}}) dx]$

e. $\int \frac{\sqrt{x}}{1+x} dx$ [Hint: Suppose $\sqrt{x} = p$]

f. $\int \frac{x}{\sqrt{x^2+4x+5}} dx$

g. $\int \frac{1}{(2x+1)\sqrt{x^2+2x+2}} dx$ [Hint: see similar example 11, page 111]

h. $\int \frac{1}{4+5\sin x} dx$

i. $\int \frac{1}{3\sin x+4\cos x} dx$

j. $\int x^4 \cos 2x dx$

k. $\int \frac{x+5}{(x+1)(x+2)^2} dx$

m. $\int \frac{2x^2+3x+4}{x^2+6x+10} dx$ [Hint: example 5, page 108]

n. $\int e^x(\tan x - \log \cos x) dx$.

"ALL THE BEST."